

Precept 3 - Solutions

Exercise 0 (Tower property)

Prove that $\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X | Y]]$.

Solution:

Let $X : \Omega \rightarrow D$ and $Y : \Omega \rightarrow D'$ be discrete random variables with a joint distribution $\mathbb{P}\{X = x | Y = y\}$ for $x \in D$ and $y \in D'$. Now,

$$\begin{aligned}\mathbf{E}(X) &= \sum_x x \mathbf{P}\{X = x\} \\ &= \sum_x x \left(\sum_y \mathbf{P}\{X = x, Y = y\} \right) \\ &= \sum_x x \left(\sum_y \mathbf{P}\{X = x | Y = y\} \mathbf{P}\{Y = y\} \right) \\ &= \sum_y \sum_x x \mathbf{P}\{X = x | Y = y\} \mathbf{P}\{Y = y\} \\ &= \sum_y \mathbf{E}(X | Y = y) \mathbf{P}\{Y = y\} \\ &= \mathbf{E}(\mathbf{E}(X | Y)).\end{aligned}$$

Exercise 1 (Miner problem)

A miner is trapped in a mine containing three doors. The first door leads to a tunnel that takes him to safety after two hours of travel. The second door leads to a tunnel that returns him to the mine after three hours of travel. The third door leads to a tunnel that returns him to his mine after five hours. Assuming that the miner is at all times equally likely to choose any of the doors, what is the expected length of time until the miner reaches safety?

Solution:

Let X denote the time until the miner reaches safety, and let Y denote the door he initially chooses. Now,

$$\begin{aligned}\mathbf{E}[X] &= \mathbf{E}[X|Y = 1]\mathbb{P}\{Y = 1\} + \mathbf{E}[X|Y = 2]\mathbb{P}\{Y = 2\} + \mathbf{E}[X|Y = 3]\mathbb{P}\{Y = 3\} \\ &= \mathbf{E}[X|Y = 1] \cdot \frac{1}{3} + \mathbf{E}[X|Y = 2] \cdot \frac{1}{3} + \mathbf{E}[X|Y = 3] \cdot \frac{1}{3}\end{aligned}$$

However,

$$\begin{aligned}\mathbb{E}[X|Y = 1] &= 2, \\ \mathbb{E}[X|Y = 2] &= 3 + \mathbb{E}[X], \\ \mathbb{E}[X|Y = 3] &= 5 + \mathbb{E}[X].\end{aligned}$$

To understand why this is correct consider, for instance, $\mathbb{E}[X|Y = 2]$, and reason as follows. If the miner chooses the second door, then he spends three hours in the tunnel and then returns to the mine. But once he returns to the mine the problem is as before, and hence his expected additional time until safety is just $\mathbb{E}[X]$. Hence $\mathbb{E}[X|Y = 2] = 3 + \mathbb{E}[X]$. The argument behind $\mathbb{E}[X|Y = 1] = 5 + \mathbb{E}[X]$ is similar. Thus,

$$\mathbb{E}[X] = 2 \cdot \frac{1}{3} + (3 + \mathbb{E}[X]) \cdot \frac{1}{3} + (5 + \mathbb{E}[X]) \cdot \frac{1}{3} \quad \text{hence } \mathbb{E}[X] = 10.$$

Exercise 2 (A dice game)

Two very good friends, Alice and Bob play the following game: Alice throws two fair, independent dice. Whenever the sum of two dice is less than 11, she throws it again —independent from the past rolls. If the sum is 12, Alice wins; and loses if the sum is 11.

- What is the probability that Alice wins "by" the n^{th} roll?
- What is the probability that Alice wins eventually?

Solution:

- Once again, before we go in, guns blazing and start attacking the problem with our probability tools, let us first extract from this story what we know:
 - The possible outcomes of the roll of two dice is $\{(1,1), (1,2), \dots, (6,6)\}$. There are 36 possible outcomes and on each throw only one of these 36 can occur. Hence, these outcomes are *mutually exclusive*.
 - When the sum of the two dice is equal to 12, Alice wins. This happens when we get the outcome $(6,6)$. Since our dice are fair, the probability of getting this outcome on each throw is $1/36$. Let us denote this set by $W := \{(6,6)\}$.
 - When the sum is equal to 11, Alice loses. This happens on these two outcomes: $(5,6)$ and $(6,5)$. Let us denote this set by $L := \{(5,6), (6,5)\}$. By the same reasoning above, the probability of getting a throw result belonging to this set is $2/36$.
 - Finally, when the sum is less than 11, no one wins and the game is replayed. The set of outcomes corresponding to this is $\{(1,1), (1,2), \dots, (5,5)\}$. It is easy to see that there are 33 outcomes in this set and thus the probability of getting a throw belonging to this set is $33/36$. Let us denote this set $R := \{(1,1), (1,2), \dots, (5,5)\}$.
 - Let W_k, L_k, R_k denote the events that the k^{th} roll lands in the sets W, L, R respectively. We are given that rolls are independent of each other. Therefore for different values of k the events W_k are independent of each other, same holds for L_k and R_k .

What does it mean for Alice to win by the n^{th} roll? In plain English, this means that she wins “at” the first game OR she wins “at” the second game OR she wins “at” the third game OR $\dots$ OR she wins “at” the n^{th} game. We see that the event “Alice wins by n^{th} roll” can be written as a union of smaller events. That is

$$\begin{aligned} \{\text{Alice wins by } n^{th} \text{ roll}\} &= \{\{\text{Alice wins “at” the first roll}\} \text{ OR} \\ &\quad \{\text{Alice wins “at” the second roll}\} \text{ OR} \\ &\quad \{\text{Alice wins “at” the third roll}\} \text{ OR} \\ &\quad \vdots \\ &\quad \{\text{Alice wins “at” the } n^{th} \text{ roll}\}\}. \end{aligned}$$

Using set notation, we replace *OR* with $\cup$ to get

$$\begin{aligned} \{\text{Alice wins by } n^{th} \text{ roll}\} &= \{\{\text{Alice wins “at” the first roll}\} \cup \\ &\quad \{\text{Alice wins “at” the second roll}\} \cup \\ &\quad \{\text{Alice wins “at” the third roll}\} \cup \\ &\quad \vdots \\ &\quad \{\text{Alice wins “at” the } n^{th} \text{ roll}\}\} \\ &= \bigcup_{k=1}^n \{\text{Alice wins “at” the } k^{th} \text{ roll}\} \end{aligned}$$

Alice winning at the k^{th} roll is equivalent to neither Bob nor Alice winning the first $k - 1$ games and then Alice winning on the k^{th} roll. But we can write this using what we defined before as follows:

$$\{\text{Alice wins “at” the } k^{th} \text{ roll}\} = \{R_1 \cap R_2 \cap \dots \cap R_{k-1} \cap W_k\}.$$

We can now introduce probabilities:

$$\begin{aligned} \mathbb{P}\{\text{Alice wins “at” the } k^{th} \text{ game}\} &= \mathbb{P}\{R_1 \cap R_2 \cap \dots \cap R_{k-1} \cap W_k\} \\ &= \mathbb{P}\{R_1\} \cdot \dots \cdot \mathbb{P}\{R_{k-1}\} \cdot \mathbb{P}\{W_k\} \\ &= \left(\frac{33}{36}\right)^{k-1} \frac{1}{36}, \end{aligned}$$

where the second equality follows from the independence of the X_k s. Recall that the event that Alice wins the game “by” the n^{th} roll can be written as

$$\bigcup_{k=1}^n \{\text{Alice wins “at” the } k^{th} \text{ roll}\}.$$

Once again, by introducing probabilities, we get

$$\begin{aligned}
& \mathbb{P}\{\text{Alice wins "by" the } n^{\text{th}} \text{ roll}\} \\
&= \mathbb{P}\left(\bigcup_{k=1}^n \{\text{Alice wins "at" the } k^{\text{th}} \text{ roll}\}\right) \\
&= \sum_{k=1}^n \mathbb{P}\{\text{Alice wins "at" the } k^{\text{th}} \text{ roll}\} \\
&= \sum_{k=1}^n \left(\frac{33}{36}\right)^{k-1} \frac{1}{36} \\
&= \frac{1}{36} \sum_{k=1}^n \left(\frac{33}{36}\right)^{k-1} \\
&= \frac{1}{36} \frac{1 - \left(\frac{33}{36}\right)^n}{1 - \left(\frac{33}{36}\right)} \quad \text{by geometric series formula} \\
&= \frac{1 - \left(\frac{11}{12}\right)^n}{3}.
\end{aligned}$$

where the second equality is from the fact that we have the union of disjoint events.

- b. The event “eventually winning the game” can be written as

$$\bigcup_{k=1}^{\infty} \{\text{Alice wins "at" the } k^{\text{th}} \text{ roll}\}.$$

So we have

$$\begin{aligned}
& \mathbb{P}\{\text{Alice eventually wins the game}\} \\
&= \frac{1}{36} \sum_{k=1}^{\infty} \left(\frac{33}{36}\right)^{k-1} \\
&= \frac{1}{36} \frac{1}{1 - \left(\frac{33}{36}\right)} \quad \text{by geometric series formula} \\
&= \frac{1}{3}.
\end{aligned}$$

Exercise 3 (Collecting action figures)

A cereal company puts an action figure in each box of cereal they sell. There are n different types of action figures that can be obtained. Each newly obtained action figure is, independently, type i with probability p_i , $i = 1, \dots, n$. Find the expected number of distinct types obtained in a collection of k cereal boxes.

Solution:

Let X be the number of distinct types in the collection of k cereal boxes. Define

$$Y_i = \begin{cases} 1 & \text{if type } i \text{ is NOT in the collection of } k \text{ cereal boxes} \\ 0 & \text{else} \end{cases}, \quad i \in \{1, \dots, n\}.$$

Note that

$$\mathbb{E}Y_i = \mathbb{P}\{Y_i = 1\} = (1 - p_i)^k.$$

(This is why we define Y_i with the “NOT”: to avoid dealing with the more complicated $1 - (1 - p_i)^k$.) Clearly, we have

$$X = n - \sum_{i=1}^n Y_i.$$

By the linearity of expectation, we have

$$\mathbb{E}X = n - \sum_{i=1}^n \mathbb{E}Y_i = n - \sum_{i=1}^n (1 - p_i)^k.$$